

Camouflaged Past, OMNIPRESENT FUTURE



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Automation is defined as carrying out a set of predefined procedures to achieve a specific outcome. Introduction of engineering automation can be dated back to the time when command-line interfaces (CLIs) were overtaken by graphical user interfaces (GUIs). Automation has passed technical barriers from engineering to medicine to commerce to entertainment.

Automation is necessitated due to the infinite amount of data being spilled over by the information age. Every data set has different specs, requirements, objectives and outcomes. Fig. 1 sheds some light on the various automations embedded in our lives.

Scripting languages like Python, TCL, PHP, Perl and PowerShell are utilised in all fields of engineering. These provide a basis for you to carry out complex tasks easily. These languages are sometimes referred to as very-high-level languages, as these are used to direct various tools and software to work.

Scripting languages are commonly interpretable rather than compilable. Primitive scripting can be seen in the form of Windows batch files (.bat), which run a set of codes sequentially to perform a task from opening MS Paint to shutting down Windows.

User interfaces inherently use scripts. A simple example to summarise scripts for automation is a single-line batch file as shown in Fig. 2. The file, when executed, copies all .png images from an external device on G:\ to C:\ images\ with one click.

In electronics, integrated circuit (IC) designs require scripts to translate designs from one technology to another. An often-used script is Tool Command Language (TCL). Timing analysis, power analysis and floor planning all require scripts to analyse the behaviour of complex circuits being cramped into smaller semiconductor dies to satisfy Gordon Moore's first law.

Embedded systems use small scripts to embed more functions, thus blurring the barriers between general-purpose systems and application-specific systems—smartphones are the best example of this transition.

With the introduction of the Internet of Things (IoT), embedded systems have entered a new era. Crunching large data (Big Data) provided by millions of IoT devices is a task simplified by Python scripting. Sensors and actuators with networking capability being put on a single board require scripts to operate. MicroPython is an example being utilised by various designers.

Fig. 1: Various automations embedded in our lives

ENGINEERING FIELDS							
Electronics		Information technology		Design engineering			Fields
Integrated circuits	Embedded systems	Testing	Programming	Automobiles	Constructions	Machines	
FPGA	The IoT	SQL	C/C++/C#	CAD/CAM			Technology
ASIC	Microcontroller	Oracle	Java				
Scripting	Scripting	Scripting	Scripting	Scripting	Scripting	Scripting	
Device prototypes	Smartphones	Apps	Operating systems	Rockets	Buildings	Earthmovers	Applications
Emulators	Wearable techs	Simulators	Websites	Aeroplanes	Architectures	Drilling rigs	